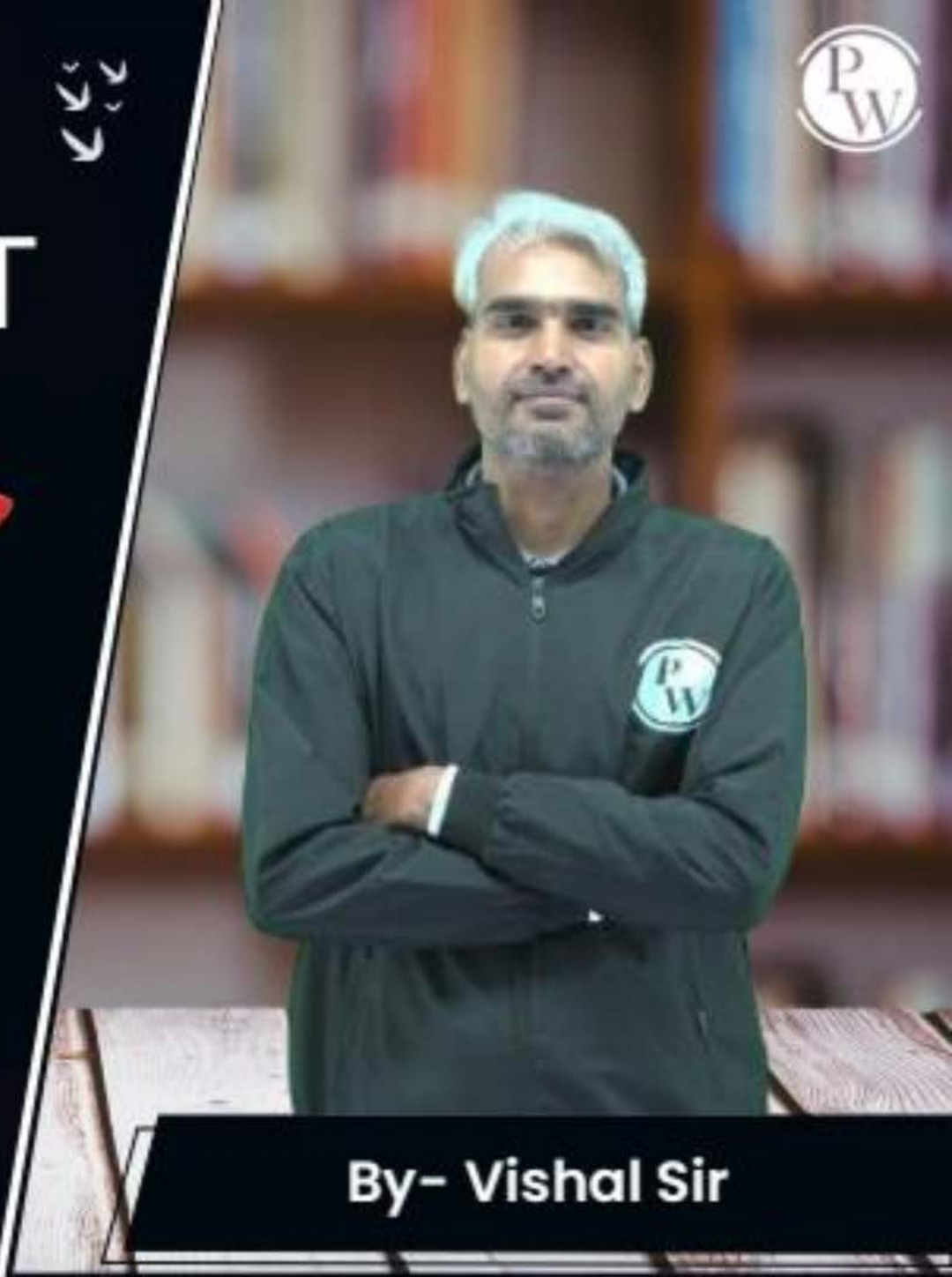
Computer Science & IT



Database Management System

Query Languages

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Execution of Nested queries
- **Topic** AS clause
- **Topic** WITH clause
- **Topic** Comparison with NULL { IS / IS NOT }

Slide

Topics to be Covered



Topic Comparison with regular expression

Topic Practice questions on SQL

Slide



Topic : Regular expression

Write the regular Expression for the names

Starting with 'D'

Ends with I

3rd letter from last is 'N'

and Contains at-least 8 letters



Topic : Regular expression

Underscore → '_' : Exactly one character

'%' : Represent 0 or more character



Topic : Regular expression

Write the regular Expression for the names

Starting with 'D'

Ends with I

3rd letter from last is 'N'

and Contains at-least 8 letters

'D,_,_,_,_,%,N,_,I'



Topic : Comparison with regular expression

To compare with regular Expression operator used is "LIKE"

Student(Sid, Sname, marks)

eg:
```
Select Sid
From Student
Where Sname LIKE 'D,%'
```

It will output the Sids of all students whose names starts with D.

Note: Complement of "LIKE" is "NOT LIKE"

#Q. Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)
Course(cNo, cName, dNo) Register(sNo, cNo)

Student

sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept

dNo	dName
D01	CSE
D02	EEE

Course

cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

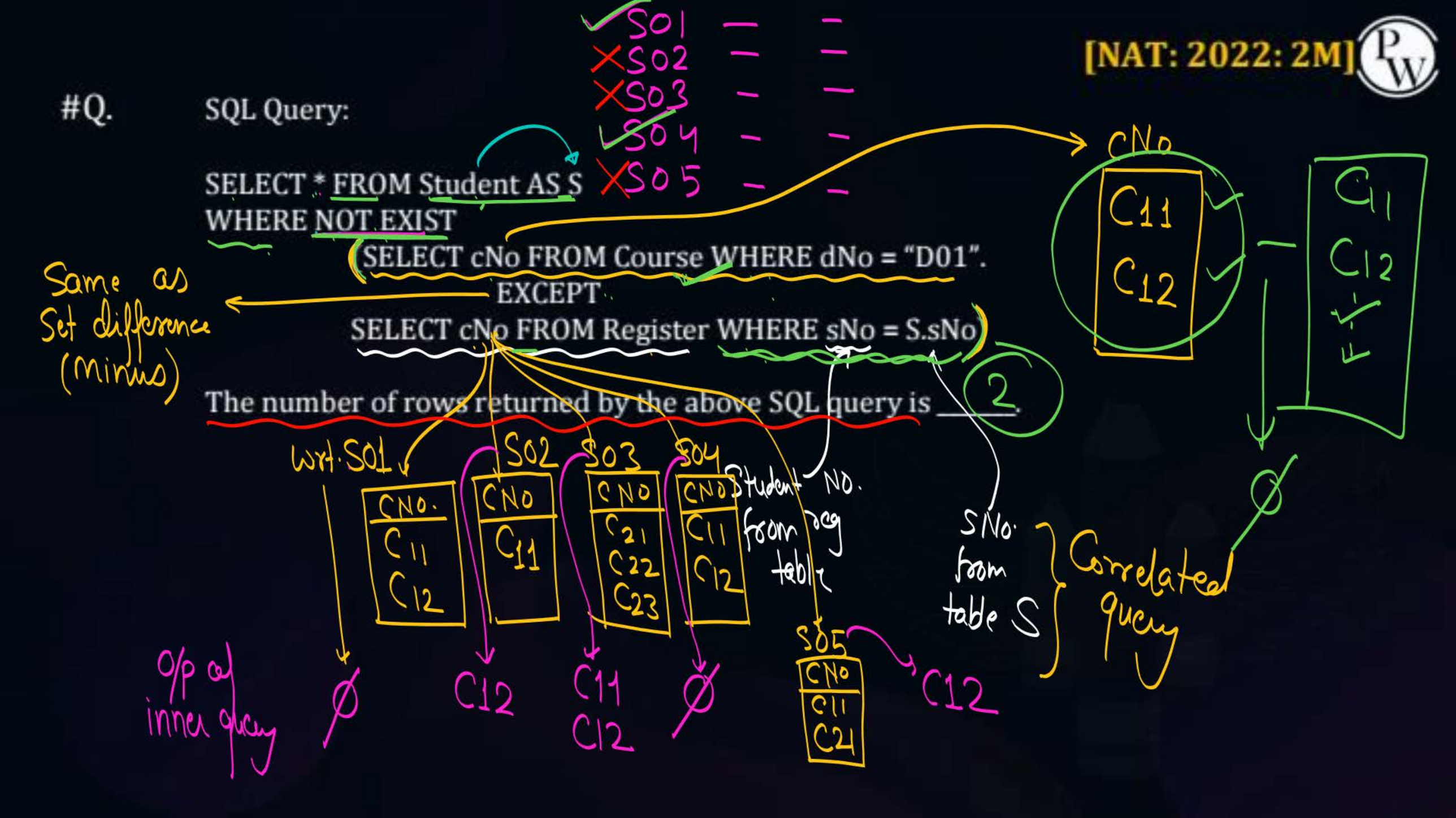
[NAT: 2022: 2M]

#Q. SQL Query:

```
SELECT * FROM Student AS S
WHERE NOT EXIST
    (SELECT cNo FROM Course WHERE dNo = "D01".
        EXCEPT
    SELECT cNo FROM Register WHERE sNo = S.sNo)
```

Same as Set difference (minus)

The number of rows returned by the above SQL query is ____.



#Q. Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)
School (school-id, sch-name, sch-address, sch-phone)
Enrolment (school-id, sch-roll-no, erollno, examname)
ExamResult (Erollno, examname, marks)

(A)for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students

(B)for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students

(C)for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam

(D)nothing; the query has a syntax error

Question Continues in Next Slide

[MCQ: 2008: 2M]

#Q. What does the following SQL query output?

```
SELECT sch-name, COUNT (*)
FROM School C, Enrolment E, ExamResult R
WHERE   E.school-id = C.school-id
                AND
        E.examname = R.examname
                AND
        E.erollno = R.erollno
                AND
        R.marks = 100
                AND
        E.school-id IN
                (SELECT school-id
                FROM student GROUP BY school-id HAVING COUNT (*) > 200)
GROUP By school-id
```

≡ C x E x R

Sid = 001

> 200

SchoolId having more than 200 students

Sch-name

#Q. [MCQ: 2009: 2M]

Consider the following relational schema:

Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
FROM Suppliers S
WHERE S.sid NOT IN
    (SELECT C.sid
     FROM Catalog C
     WHERE C.pid NOT IN
         (SELECT P.pid
          FROM Parts P
          WHERE P.color<> \'blue\'))
```

Suppliers who did not supply any part OR Supplied only non-blue Color parts.

Sids of suppliers who have supplied at least one blue Color part

{P1}

Pid of non-blue Color Parts

Not equal

S_1	P_1
S_1	P_2
S_2	P_1
S_3	P_2

P_1	R
P_2	B

Question Continues in Next Slide

#Q. Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. Find the names of all suppliers who have supplied a non-blue part.

{ Suppliers who have supplied at least one non-blue Part }

B. Find the names of all suppliers who have not supplied a non-blue part.

{ Suppliers who did not supply any non-blue Part }

C. Find the names of all suppliers who have supplied only blue parts.

{ Suppliers who have supplied only blue parts }

D. Find the names of all suppliers who have not supplied only blue parts.

Suppliers who have supplied some other Color Parts along with some blue Color part

E: None of the above



2 mins Summary

Topic Comparison with regular expression

Topic Practice questions on SQL

Slide

THANK - YOU

Slide